

WATERWALKER · REQUIREMENTS

WaterWalker requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/waterwalker		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWalker requirements

These are first-pass requirements for the concept. Targets are desk proposals for review and must be revised from co-design sessions with the intended users (WWK-PRB-001) before the design is frozen. They will be checked by calculation at TRL 3. Concept estimates against each target are in WWK-PRC-001.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Carry a household's water in one trip	80 L of water in four 20 L jerrycans; rated gross payload 90 kg including containers; also accept 20 L clay pots	Cradle layout in the parametric model; later load test
R2	Easy to push on firm ground	Sustained push force 60 N or less at rated load on a firm, level dirt path	Rolling resistance calculation; later force gauge on a field path
R3	Usable on loose sand	Sustained push force 150 N or less at rated load on loose sand, with assist if fitted	Calculation from sinkage and rolling resistance; later field test
R4	Climb and descend slopes	Climb a 10 % grade on firm ground with a push force of 180 N or less for up to 100 m; descend a 10 % grade under control with the service brake	Force calculation; brake sizing
R5	Turn without lifting	180 degree turn within a 4.0 m diameter circle without lifting or skidding a wheel; reverse by pulling back	Geometry of the parametric model
R6	Fit narrow paths and gates	Overall width 900 mm or less	Parametric model
R7	Walk-in fit for all users	No step-over at the rear entry (50 mm or less); clear walking width 600 mm or more; hip bar height adjustable 850 to 1,050 mm; skirt guards on both rear wheels	Parametric model; later co-design fitting sessions
R8	Easy loading	Containers lifted no higher than 450 mm above the ground; loadable from either side without removing parts	Parametric model
R9	Light enough to handle empty	Empty mass 35 kg or less without assist, 42 kg or less with assist	Mass roll-up from the parametric model
R10	Safe braking and parking	Service brake on both rear wheels operable while walking; parking brake holds the rated gross mass on a 20 % grade	Brake force calculation
R11	Affordable prototype	Parts for one prototype \$450 or less (<code>project.yaml</code> budget)	Priced BOM
R12	Repairable in a rural market	All wear parts (tires, tubes, spokes, bearings, cables, brake shoes) are standard 26 in bicycle parts; mild steel frame repairable by a local welder; only common hand and bicycle tools needed	Design review of the BOM; later review with a local mechanic

■ Assumptions

- A 20 L jerrycan weighs about 1.1 kg empty and measures about 360 x 175 x 430 mm. A 20 L clay pot is assumed to be about 380 mm in diameter and 5 to 8 kg empty.
- Sustained push forces are set with reference to published manual-handling guidance, which suggests roughly 100 to 150 N for sustained pushing by women over long distances and more for short efforts. The limits must be confirmed with the users, including girls and older women.
- A household of five uses about 100 L per day (about 20 L per person).
- R5 allows a larger circle than a compact rollator because the concept needs a long wheelbase to keep the swivelling front wheels clear of the cradle (WWK-PRC-001). A three-point turn is expected on paths narrower than 4 m.